وزارة الصحة العراقية
Iraqi Ministry of Health
Founded 1920 تأسست

دائرة الصحة العامة
Public Health Directorate

2025

VOLUME 2

NATIONAL GUIDELINES FOR PRIMARY HEALTH CARE PHYSICIANS

Chronic Respiratory Diseases.(CRDs)

Management of Chronic Obstructive Pulmonary Diseases



Directorate of Public Health
Non-Communicable Disease Department

IRAQ NCD

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National Guidelines for Primary Health Care Physicians

Guidelines for the Management of Chronic Obstructive Pulmonary Diseases For Primary Health Care

Directorate of Public Health

Non-Communicable Disease Department

2025

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Abbreviation	Terminology
AATD	Alpha-1 Antitrypsin Deficiency
BMI	Body Mass Index
CAT	Copd Assessment Tool
CCQ	Clinical Copd Questionnaire
COPD	Chronic Obstructive Pulmonary Disease
CXR	Chest X- Ray
ETS	Environmental Tobacco Smoke
FEV1	Forced Expiratory Volume in One Second
FVC	Forced Vital Capacity
ICS	Inhaled Corticosteroid
JVP	Jugular Venous Pressure
LABA	Long-Acting Beta-2-Agonists
LTOT	Long-Term Oxygen Therapy
MRC	Medical Research Council
NICE	National Institution for Health And Care Excellence
pMDI	Pressurised Metered Dose Inhaler
SABA	Short -Acting Beta-2-Agonists
SAMA	Long-Acting Anti-Muscarinic Antagonist
SBOT	Short-Burst Oxygen Therapy
WHO	World Health Origination

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Chronic Obstructive Pulmonary Disease

Introduction

Chronic Obstructive Pulmonary Disease (COPD) is now one of the top three causes of death worldwide and 90% of these deaths occur in low- and middle-income countries (LMICs). More than 3 million people died of COPD in 2012accounting for 6% of all deaths globally. COPD represents an important public health challenge that is both preventableand treatable. COPD is a major cause of chronic morbidity and mortality throughout the world; many people suffer from this disease for years and die prematurely from it or its complications. Globally, the COPD burden is projected to increase incoming decades because of continued exposure to COPD risk factors and aging of the population. COPD is a common, preventable, and treatable disease, but extensive under-diagnosis and misdiagnosis leads to patients receiving no treatment or incorrect treatment. Appropriate and earlier diagnosis of COPD can have a very significant public-health impact.

Prevalence

It is estimated that **the global prevalence of COPD is 10.3%,** With the increasing prevalence of smoking and aging populations the prevalence of COPD is expected to rise.

Morbidity and Mortality

Morbidity measures traditionally include physician visits, emergency department visits, and hospitalizations. morbidity due to COPD increases with age, and in patients with COPD the development of comorbidities are seen at an earlier age. Morbidity in COPD may also be influenced by concomitant chronic conditions (e.g., cardiovascular disease, musculoskeletal impairment, diabetes mellitus) that are related to smoking, aging and/or COPD.it can be estimated that **globally there are around three million deaths annually due to COPD.**

Pathogenesis

COPD is the end-result of complex, cumulative and dynamic gene-environment interactions over the lifetime that can damage the lungs and/or alter their normal developmental or aging processes (GETomics). Understanding the relationships and interactions between the genetic (G) background of the host and varied environmental (E) risk factors over the lifetime(T).

Definition of COPD

Chronic Obstructive Pulmonary Disease (COPD) is a heterogeneous lung condition characterized by chronic respiratory symptoms (dyspnea, cough, sputum production and/or exacerbations) due to abnormalities of the airways (bronchitis, bronchiolitis) and/or alveoli (emphysema) that cause persistent, often progressive, airflow obstruction.

Causes and Risk Factors

- **COPD results** from gene (G)-environment(E) interactions occurring over the lifetime (T) of the individual (GETomics) that can damage the lungs and/or alter their normal development/aging processes.

- **Environmental risk factors:** The main environmental exposures leading to COPD are:

  - ✓ Cigarette smoking
  - ✓ Biomass exposure
  - ✓ Occupational exposures
  - ✓ Air pollution

- **Genetic factors**

The genetic risk factor for COPD are mutations in the SERPINA1 gene that leads to the hereditary deficiency of α-1 antitrypsin (AATD)

- **Asthma and airway hyper-reactivity**

Asthma may also be a risk factor for the development of chronic airflow obstruction and COPD.

- ✓ Infections
- ✓ Sex
- ✓ Race and ethnicity
- ✓ Socioeconomic status

Proposed Taxonomy (Etiotypes) for COPD

Classification	Description
Genetically determined COPD (COPD-G)	Alpha-1 antitrypsin deficiency (AATD) Other genetic variants with smaller effects acting in combination
COPD due to abnormal lung development (COPD-D)	Early life events, including premature birth and low birthweight, among others
Environmental COPD	
Cigarette smoking COPD (COPD-C)	• Exposure to tobacco smoke, including *in utero* or via passive smoking • Vaping or e-cigarette use • Cannabis
Biomass and pollution exposure COPD (COPD-P)	Exposure to household pollution, ambient air pollution, wildfire smoke, occupational hazards
COPD due to infections (COPD-I)	Childhood infections, tuberculosis-associated COPD, HIV-associated COPD
COPD & asthma (COPD-A)	Particularly childhood asthma
COPD of unknown cause (COPD-U)	

*Adapted from Celli et al. (2022) and Stolz et al. (2022)

- **Pathophysiology:**
  - ✓ Airflow obstruction and gas trapping
  - ✓ Hyperinflation
  - ✓ Pulmonary gas exchange abnormalities
  - ✓ Pulmonary hypertension
  - ✓ Exacerbations
  - ✓ Multimorbidity

Diagnosis and assessment

Clinical presentation:

Symptoms

Chronic dyspnea is the most characteristic symptom of COPD. Cough with sputum production is present in up to 30%of patients. These symptoms may vary from day-to-day and may precede the development of airflow obstruction by many years. Individuals, particularly those with COPD risk factors, presenting with these symptoms should be examined to search for the underlying cause(s). Airflow obstruction may also be present without chronic dyspnea and/or cough and sputum production and *vice versa*.

Clinical Indicators for Considering a Diagnosis of COPD

Consider the diagnosis of COPD, and perform spirometry, if any of these clinical indicators are present:
(these indicators are not diagnostic themselves, but the presence of multiple key indicators increases the probability of the presence of COPD; in any case, spirometry is required to establish a diagnosis of COPD)

Dyspnea that is	Progressive over time Worse with exercise Persistent
Recurrent wheeze	
Chronic cough	May be intermittent and may be non-productive
Recurrent lower respiratory tract infections	
History of risk factors	Tobacco smoke (including popular local preparations) Smoke from home cooking and heating fuels Occupational dusts, vapors, fumes, gases and other chemicals Host factors (e.g., genetic factors, developmental abnormalities, low birthweight, prematurity, childhood respiratory infections etc.)

Dyspnea

Dyspnea is a cardinal symptom of COPD and a major cause of the disability and anxiety associated with the disease. Dyspnea comprises a sensory and an affective component. Typically, COPD patients describe their dyspnea as a sense of increased effort to breathe, chest heaviness, air hunger, or gasping. However, the terms used to describe dyspnea may vary both individually and culturally.

Dyspnea is highly prevalent across all stages of airflow obstruction. It occurs particularly during exertion or physical activity. Moderate-to-severe dyspnea has been reported by > 40% of patients diagnosed with COPD in primary care.

Chronic cough

Chronic cough is often the first symptom of COPD and is frequently discounted by the patient as an expected consequence of smoking and /or environmental exposures. Initially, the cough may be intermittent, but subsequently it may be present every day, often throughout the day. Chronic cough in COPD may be productive or non-productive. In some cases, significant airflow obstruction may develop without the presence of a cough. Syncope during cough in patients with severe COPD can occur due to rapid increases in intrathoracic pressure during prolonged attacks of coughing. Coughing spells may also cause rib fractures, which are sometimes asymptomatic.

Other Causes of Chronic Cough

INTRATHORACIC	EXTRATHORACIC
• Asthma	• Chronic Allergic Rhinitis
• Lung Cancer	• Post Nasal Drip Syndrome (PNDS)
• Tuberculosis	• Upper Airway Cough Syndrome (UACS)
• Bronchiectasis	• Gastroesophageal Reflux
• Left Heart Failure	• Medication (e.g., ACE Inhibitors)
• Interstitial Lung Disease	
• Cystic Fibrosis	
• Idiopathic Cough	

Sputum production

COPD patients commonly raise small quantities of tenacious sputum with coughing. Regular production of sputum forthree or more months in two consecutive years (in the absence of any other conditions that may explain it) is the classical definition of chronic bronchitis. Sputum production is often difficult to evaluate because patients may swallow sputum rather than expectorate it, a habit that is subject to significant cultural and sex variation. Furthermore, sputum production can be intermittent with periods of flare-up interspersed with periods of remission. Patients producing large volumes of sputum may have underlying bronchiectasis. The presence of purulent sputum reflects an increase in inflammatory mediators, and may identify the onset of a bacterial exacerbation, though the association is relatively weak.

Wheezing and chest tightness

Inspiratory and/or expiratory wheezes and chest tightness are symptoms that may vary between days, and over the course of a single day. Alternatively, widespread inspiratory or expiratory wheezes can be present on auscultation. An absence of wheezing or chest tightness does not exclude a diagnosis of COPD, nor does the presence of these symptoms confirm a diagnosis of asthma.

Fatigue

Fatigue is the subjective feeling of tiredness or exhaustion and is one of the most common and distressing symptoms experienced by people with COPD. People with COPD describe their fatigue as a feeling of "general tiredness" or as a feeling of being "drained of energy". Fatigue impacts a patient's ability to perform activities of daily living and their quality of life.

Additional clinical features in severe disease

Weight loss, muscle mass loss, and anorexia are common problems in patients with severe and very severe COPD. They have prognostic importance and can also be a sign of other diseases, such as tuberculosis or lung cancer, and therefore should always be investigated. Ankle swelling may indicate the presence of cor pulmonale. Symptoms of depression and/or anxiety merit specific enquiry when obtaining the medical history because they are common in COPD, are associated with poorer health status, increased risk of exacerbations, and emergency hospital admission, and are treatable.

Medical history

A detailed medical history of a new patient who is known, or suspected, to have COPD should include:

- *Patient's exposure to risk factors*, such as smoking and environmental exposures (household/outdoor).
- *Past medical history*, including early life events (prematurity, low birth weight, maternal smoking during pregnancy, passive smoking exposure during infancy), asthma, allergy, sinusitis, or nasal polyps; respiratory infections in childhood; HIV; tuberculosis.
- *Family history of COPD or other chronic respiratory disease.*
- *Pattern of symptom development*: COPD typically develops in adult life and most patients are conscious of increased breathlessness, more frequent or prolonged "winter colds," and some social restriction for a number of years before seeking medical help.
- *History of exacerbations or previous hospitalizations for respiratory disorder.* Patients may be aware of periodic worsening of symptoms even if these episodes have not been identified as exacerbations of COPD.
- *Presence of comorbidities*, such as heart disease, osteoporosis, musculoskeletal disorders, anxiety and depression, and malignancies that may also contribute to restriction of activity.
- *Impact of disease on patient's life*, including limitation of activity, missed work and economic impact, effect on family routines, feelings of depression or anxiety, wellbeing, and sexual activity.
- *Social and family support available to the patient.*
- *Possibilities for reducing risk factors, especially smoking cessation.*

Physical examination

Although an important part of patient care, a physical examination is rarely (if ever) diagnostic in COPD. Physical signs of airflow obstruction are usually not present until significant impairment of lung function has occurred, and detection based on physical examination has relatively low sensitivity and specificity. A number of physical signs (e.g., lung hyperinflation, cyanosis) may be present in COPD, but their absence does not exclude the diagnosis.

Differential Diagnosis of COPD

Diagnosis	Suggestive Features
COPD	Symptoms slowly progressive History of tobacco smoking or other risk factors
Asthma	Variable airflow obstruction Symptoms vary widely from day to day Symptoms worse at night/early morning Allergy, rhinitis, and/or eczema also present Often occurs in children Family history of asthma
Congestive heart failure	Chest X-ray shows dilated heart, pulmonary edema Pulmonary function tests indicate volume restriction, not airflow obstruction
Bronchiectasis	Large volumes of purulent sputum Commonly associated with bacterial infection Chest X-ray/HRCT shows bronchial dilation
Tuberculosis	Onset at all ages Chest X-ray shows lung infiltrate Microbiological confirmation High local prevalence of tuberculosis
Obliterative bronchiolitis	Can occur in children Seen after lung or bone marrow transplantation HRCT on expiration shows hypodense areas
Diffuse panbronchiolitis	Predominantly seen in patients of Asian descent Most patients are male and nonsmokers Almost all have chronic sinusitis Chest X-ray & HRCT show diffuse small centrilobular nodular opacities & hyperinflation

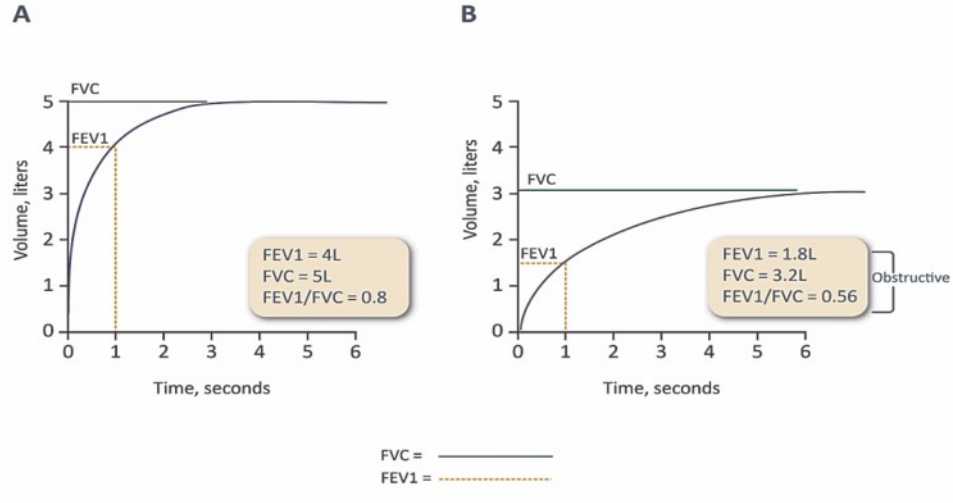
These features tend to be characteristic of the respective diseases, but are not mandatory. For example, a person who has never smoked may develop COPD (especially in LMICs where other risk factors may be more important than cigarette smoking).

Spirometry

Forced spirometry is the most reproducible and objective measurement of airflow obstruction. It is a noninvasive, reproducible, cheap, and readily available test. Good quality spirometric measurement is possible in any healthcare setting and all healthcare workers who care for people with COPD should have access to spirometry.

peak expiratory flow measurement alone cannot be reliably used as the only diagnostic test because of its weak specificity.The spirometric criterion for airflow obstruction selected by GOLD for the diagnosis of COPD remains a post- bronchodilator ratio of FEV1/FVC < 0.7. This criterion is simple and independent of reference values because it relatesto variables measured in the same individual.

A. Spirometry - Normal Trace **B. Spirometry - Airflow Obstruction**



Role of Spirometry in COPD

- **Diagnosis**
- **Assessment of severity of airflow obstruction (for prognosis)**
- **Follow-up assessment**
  - Therapeutic decisions
    - Pharmacological in selected circumstances (e.g., discrepancy between spirometry and level of symptoms)
    - Consider alternative diagnoses when symptoms are disproportionate to degree of airflow obstruction
    - Non-pharmacological (e.g., interventional procedures)
  - Identification of rapid decline

Initial assessment

Once the diagnosis of COPD has been confirmed by spirometry, in order to guide therapy COPD assessment must focuson determining the following five fundamental aspects:

- Severity of airflow obstruction
- Nature and magnitude of current symptoms
- Previous history of moderate and severe exacerbations
- Blood eosinophil count
- Presence and type of other diseases (multimorbidity)

Severity of airflow obstruction

In the presence of FEV1/FVC ratio < 0.7 the assessment of **airflow obstruction severity** in COPD (note that this may be different from severity of the *disease*) is based on the post-bronchodilator value of FEV1 (% reference).

GOLD Grades and Severity of Airflow Obstruction in COPD (based on post-bronchodilator FEV1)

In COPD patients (FEV1/FVC < 0.7):

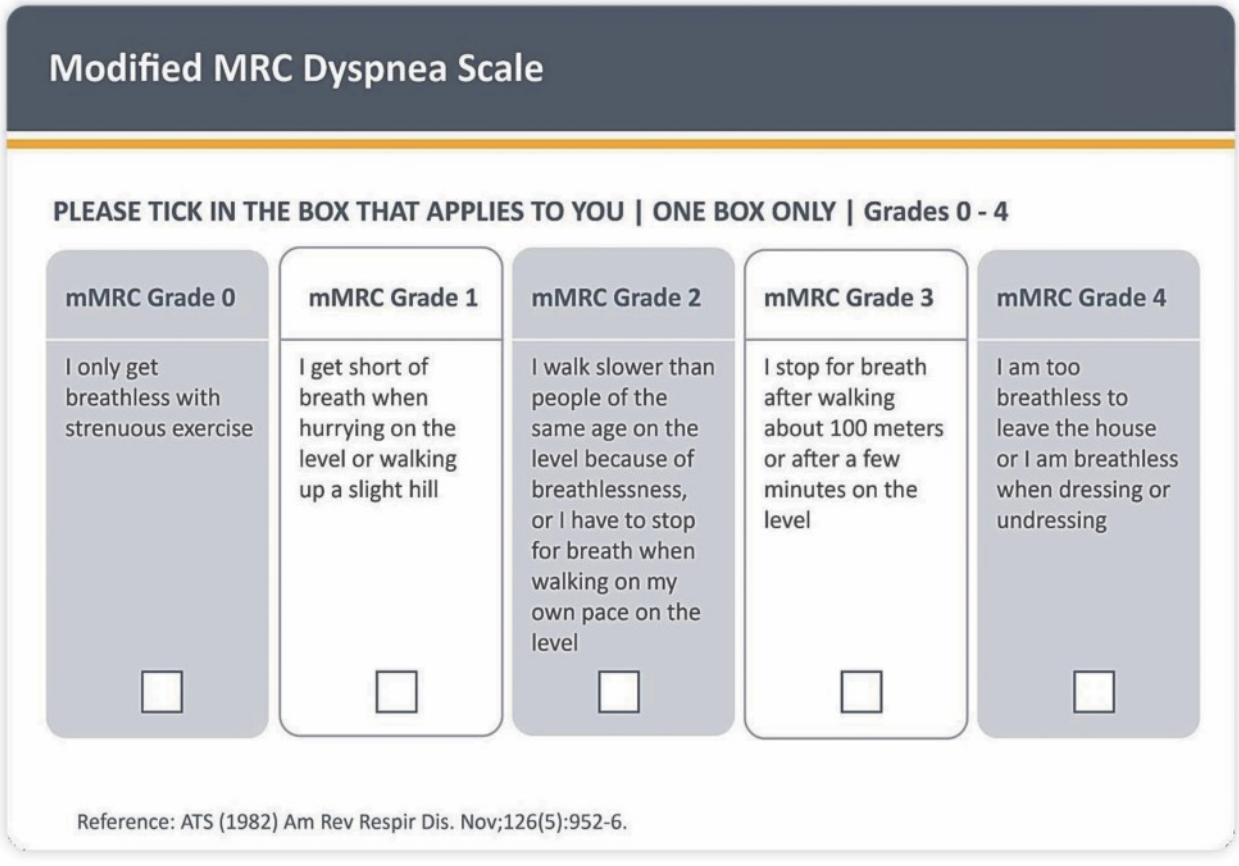
GOLD 1:	Mild	FEV1 ≥ 80% predicted
GOLD 2:	Moderate	50% ≤ FEV1 < 80% predicted
GOLD 3:	Severe	30% ≤ FEV1 < 50% predicted
GOLD 4:	Very Severe	FEV1 < 30% predicted

Symptoms

Because there is only a weak correlation between the severity of airflow obstruction and the symptoms experienced by the patient or the impairment of their health status, formal assessment of symptoms using validated questionnaires is required.

Dyspnea questionnaire: the modified Medical Research Council (mMRC) dyspnea scale

The mMRC scale was the first questionnaire developed to measure breathlessness, which is a key symptom in many patients with COPD, although often unrecognized. The mMRC score relates well to other multidimensional health status measures and predicts future mortality risk.



Multidimensional questionnaires

It is now recognized that COPD impacts patients beyond dyspnea. For this reason, multidimensional questionnaires are recommended.The CAT is an 8-item questionnaire that assesses health status in patients with COPD. It was developed to be applicable worldwide and validated translations are available in a wide range of languages.

CAT™ Assessment

For each item below, place a mark (x) in the box that best describes you currently.
Be sure to only select one response for each question.

EXAMPLE: I am very happy	0 X 2 3 4 5	I am very sad	Score
I never cough	0 1 2 3 4 5	I cough all the time	
I have no phlegm (mucus) in my chest at all	0 1 2 3 4 5	My chest is completely full of phlegm (mucus)	
My chest does not feel tight at all	0 1 2 3 4 5	My chest feels very tight	
When I walk up a hill or one flight of stairs I am not breathless	0 1 2 3 4 5	When I walk up a hill or one flight of stairs I am very breathless	
I am not limited doing any activities at home	0 1 2 3 4 5	I am very limited doing activities at home	
I am confident leaving my home despite my lung condition	0 1 2 3 4 5	I am not at all confident leaving my home because of my lung condition	
I sleep soundly	0 1 2 3 4 5	I don't sleep soundly because of my lung condition	
I have lots of energy	0 1 2 3 4 5	I have no energy at all	

Reference: Jones et al. ERJ 2009; 34 (3); 648-54.

TOTAL SCORE: ☐

Blood eosinophil count

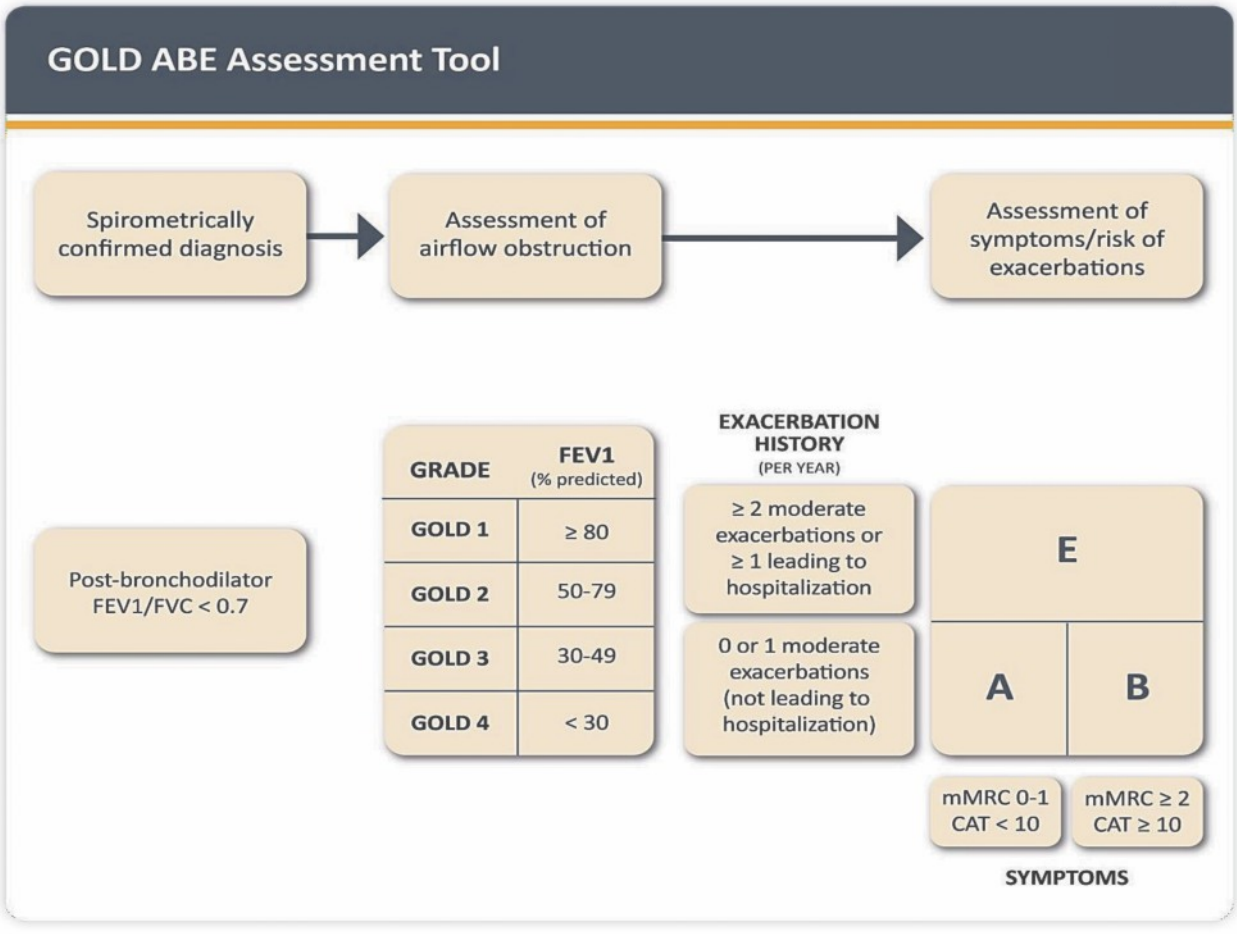
A number of studies have shown that blood eosinophil counts predict the magnitude of the effect of ICS (added on top of regular maintenance bronchodilator treatment) in preventing future exacerbations and blood eosinophil counts are recommended by GOLD to guide the use of ICS as part of pharmacological management.

Multimorbidity

People with COPD often suffer other concomitant chronic diseases (multimorbidity). This can occur in patients with mild, moderate or severe airflow obstruction. Multimorbidity influences mortality and hospitalizations independently of the severity of airflow obstruction, and deserves specific treatment. Therefore, comorbid conditions should be

looked for routinely, and treated appropriately if present, in any patient with COPD. Recommendations for the diagnosis, assessment of severity, and management of individual comorbid diseases are thesame as for patients without COPD.

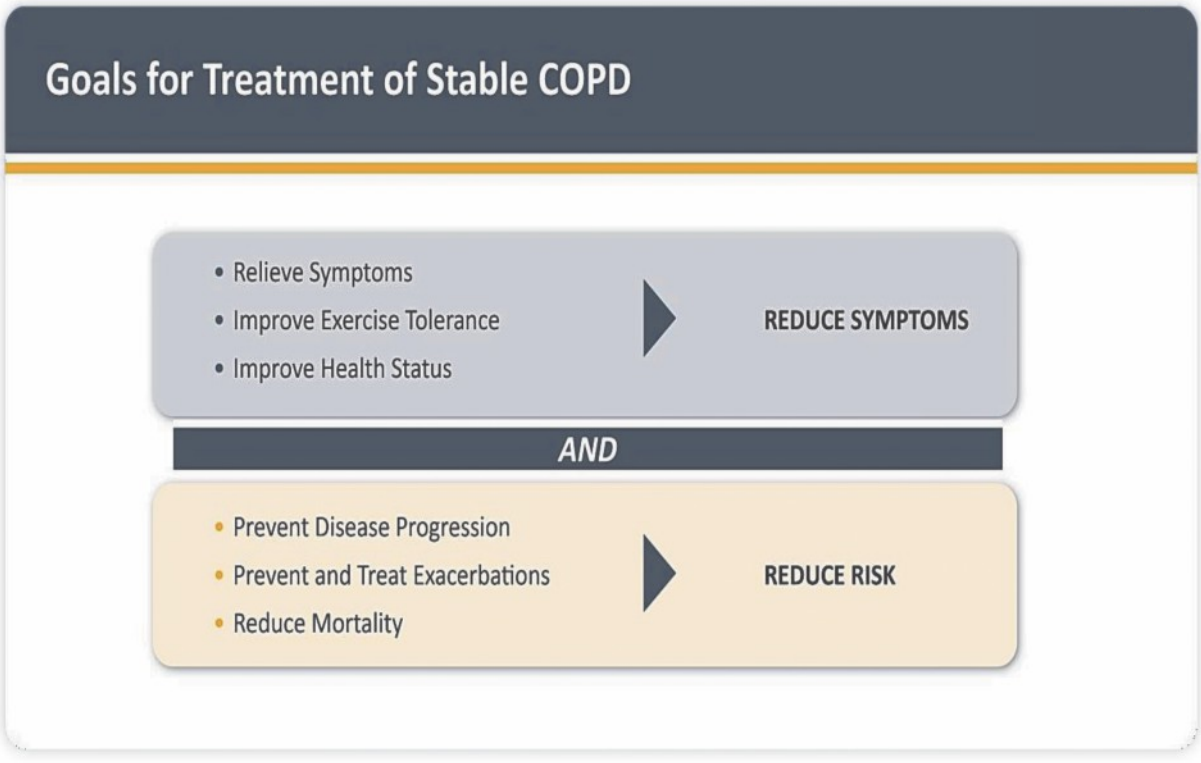
Frequent multimorbid diseases in COPD include cardiovascular disease, metabolic syndrome, osteoporosis, depression and anxiety, likely in relation to shared risk factors (e.g., aging, smoking, alcohol, diet and inactivity). Besides, COPD itself may increase the risk for other comorbid diseases (e.g., COPD (particularly emphysema) and lungcancer). Whether the association between COPD and lung cancer is due to common risk factors (e.g., smoking), involvement of shared susceptibility genes and/or impaired clearance of carcinogens is unclear. COPD can also have significant extrapulmonary (systemic) effects including weight loss, nutritional abnormalities, and skeletal muscle dysfunction. The latter is characterized by both sarcopenia (loss of muscle cells) and abnormal function of the remaining cells. Its causes are likely multifactorial (e.g., inactivity, poor diet, inflammation and/or hypoxia) and it can contribute to exercise intolerance and poor health status in patients with COPD. Importantly, skeletal muscle dysfunction is a modifiable source of exercise intolerance by rehabilitation.



Prevention and management of COPD

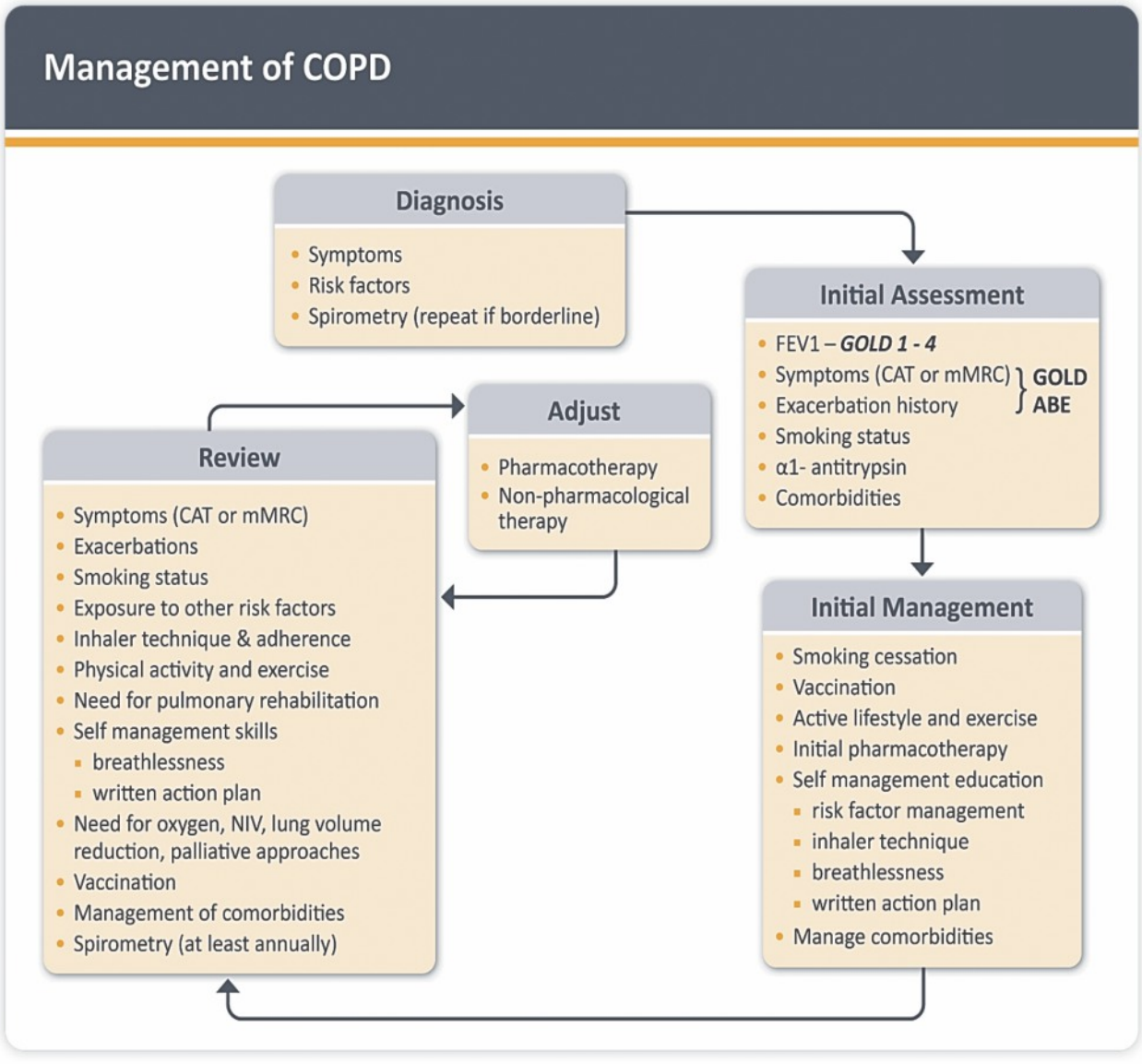
Introduction

The main treatment goals are to reduce symptoms and future risk of exacerbations. The management strategy of stable COPD should be predominantly based on the assessment of symptoms and the history of exacerbations.



Identify and reduced exposure to risk factors

Identification and reduction of exposure to risk factors is important not only for the primary prevention of COPD but also as part of the management of a COPD patient. Cigarette smoking is the most commonly encountered and easily identifiable risk factor for COPD, and smoking cessation should be continually encouraged for all individuals who smoke. Reduction of total personal exposure to occupational dusts, fumes, and gases, and to household and outdoorair pollutants, should also be addressed.



Smoking cessation

Smoking cessation is a key intervention for all COPD patients who continue to smoke. Smoking cessation has the greatest capacity to influence the natural history of COPD, it also improvesdaily symptoms, and decreases the frequency of exacerbations.

All individuals who smoke should be strongly encouraged and supported to quit. Nicotine replacement and pharmacotherapy reliably increase long-term smoking abstinence rates.

Brief Strategies to Help the Patient Willing to Quit

ASK	Systematically identify all tobacco users at every visit *Implement an office-wide system that ensures that, for EVERY patient at EVERY clinic visit, tobacco-use status is queried and documented*
ADVISE	Strongly urge all tobacco users to quit *In a clear, strong, and personalized manner, urge every tobacco user to quit*
ASSESS	Determine willingness and rationale of patient's desire to make a quit attempt. *Ask every tobacco user if he or she is willing to make a quit attempt at this time (e.g., within the next 30 days)*
ASSIST	Aid the patient in quitting *Help the patient with a quit plan; provide practical counseling; provide intra-treatment social support; help the patient obtain extra-treatment social support; recommend use of approved pharmacotherapy except in special circumstances; provide supplementary materials*
ARRANGE	Schedule follow-up contact *Schedule follow-up contact, either in person or via telephone*

Vaccinations

People with COPD should receive all recommended vaccinations in line with the relevant local guidelines.

Vaccination for Stable COPD

- Influenza vaccination is recommended for people with COPD **(Evidence B)**
- The WHO and CDC recommends SARS-CoV-2 (COVID-19) vaccination for people with COPD **(Evidence B)**
- The CDC recommends one dose of 20-valent pneumococcal conjugate vaccine (PCV20); or one dose of 15-valent pneumococcal conjugate vaccine (PCV15) followed by 23-valent pneumococcal polysaccharide vaccine (PPSV23) for people with COPD **(Evidence B)**
- Pneumococcal vaccination has been shown to reduce the incidence of community-acquired pneumonia and exacerbations for people with COPD **(Evidence B)**
- The CDC recommends the new respiratory syncytial virus (RSV) vaccine for individuals over 60 years and/or with chronic heart or lung disease **(Evidence A)**
- The CDC recommends Tdap (dTaP/dTPa) vaccination to protect against pertussis (whooping cough) for people with COPD that were not vaccinated in adolescence **(Evidence B)**, and Zoster vaccine to protect against shingles for people with COPD over 50 years **(Evidence B)**

Household and outdoor air pollution

Reducing exposure to household and outdoor air pollution requires a combination of public policy, local and national resources, cultural changes, and protective steps taken by individual patients. Reduction of exposure to smoke from biomass fuel is a crucial goal to reduce the prevalence of COPD worldwide. Efficient ventilation, non-polluting cooking stoves and similar interventions are feasible and should be recommended.

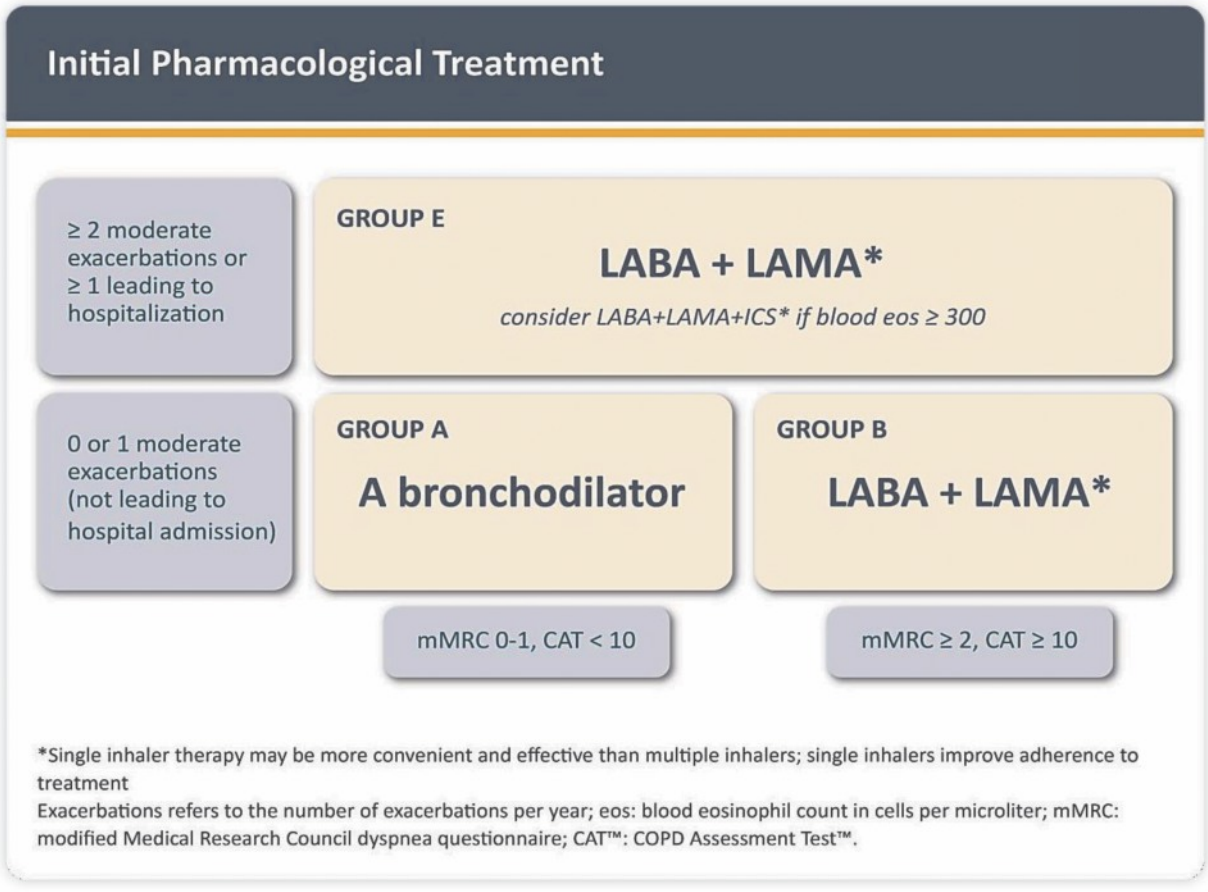
Pharmacological treatment of stable COPD

Each pharmacological treatment regimen should be individualized and guided by the severity of symptoms, risk of exacerbations, side-effects, comorbidities, drug availability and cost, and the patient's response, preference, and ability to use various drug delivery devices.

Patients should be reviewed after a suitable interval (shorter in more severe patients and longer in less severe patients) and their current level of symptoms (using either the CAT or mMRC scores) and exacerbation frequency assessed. The effect of treatment and possible adverse effects should be evaluated, and comorbidities reassessed.

Inhaler technique, adherence to prescribed therapy (both pharmacological and non-pharmacological), smoking status and continued exposure to risk factors should be checked at each clinical visit. Physical activity should be encouraged and referral for pulmonary rehabilitation considered in severe patients. The need for oxygen therapy, non-invasive ventilatory support, lung volume reduction and palliative approaches should also be considered individually, and the action plan should be updated accordingly. Spirometry should be repeated at least annually. If the patient is already receiving bronchodilator treatment, the latter should not be interrupted for performing spirometry.
Asthma and COPD may coexist in an individual patient. If a concurrent diagnosis of asthma is suspected, pharmacotherapy should primarily follow asthma guidelines.

Algorithms for the initiation and follow-up of pharmacological treatment



Group A

- All Group A patients should be offered bronchodilator treatment based on its effect on breathlessness. This can be either a short- or a long-acting bronchodilator. If available and affordable a long-acting bronchodilator is the preferred choice except in patients with very occasional breathlessness.
- This should be continued if benefit is documented.

Group B

- Treatment should be initiated with a LABA+LAMA combination. It has been shown in a RCT that in patients with ≤ 1 moderate exacerbation in the year before the study and a CAT™ ≥ 10 LABA+LAMA is superior to a LAMA with regard to several endpoints. Therefore, providing there are no issues regarding availability, cost and side-effects LABA+LAMA is the recommended initial pharmacological choice.
- If a LABA+LAMA combination is not considered appropriate, there is no evidence to recommend one class of long-acting bronchodilators over another (LABA or LAMA) for initial relief of symptoms in this group of patients. In the individual patient, the choice should depend on the patient's perception of symptom relief.

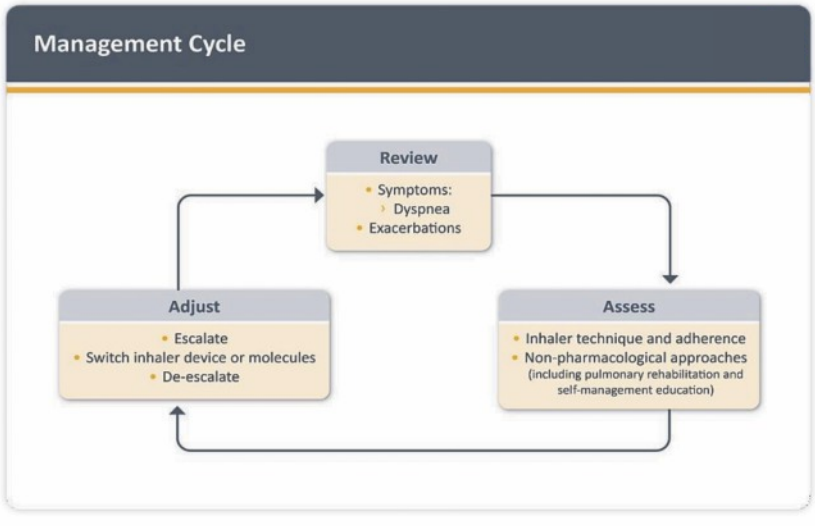
- Group B patients are likely to have comorbidities that may add to their symptomatology and impact their prognosis, and these possibilities should be investigated and treated, if present, by following national and international guidelines.

Group E

- LABA+LAMA combination was the highest ranked treatment group to reduce COPD exacerbations. Therefore, provided there are no issues regarding availability, cost and side-effects LABA+LAMA is the preferred choice for initial therapy in group E patients.
- Use of LABA+ICS in COPD is not encouraged. If there is an indication for an ICS, then LABA+LAMA+ICS has been shown to be superior to LABA+ICS and is therefore the preferred choice.
- Consider LABA+LAMA+ICS in group E if eos $\geq$ 300 cells/µL, the effect of ICS on exacerbation prevention is correlated to blood eosinophil count.
- If patients with COPD have concomitant asthma they should be treated like patients with asthma. Under these circumstances the use of an ICS is mandatory.
- Rescue short-acting bronchodilators should be prescribed to all patients for immediate symptom relief.
- Following implementation of therapy, patients should be reassessed for attainment of treatment goals and identification of any barriers for successful treatment . Following review of the patient's response to treatment initiation, adjustments may be needed.

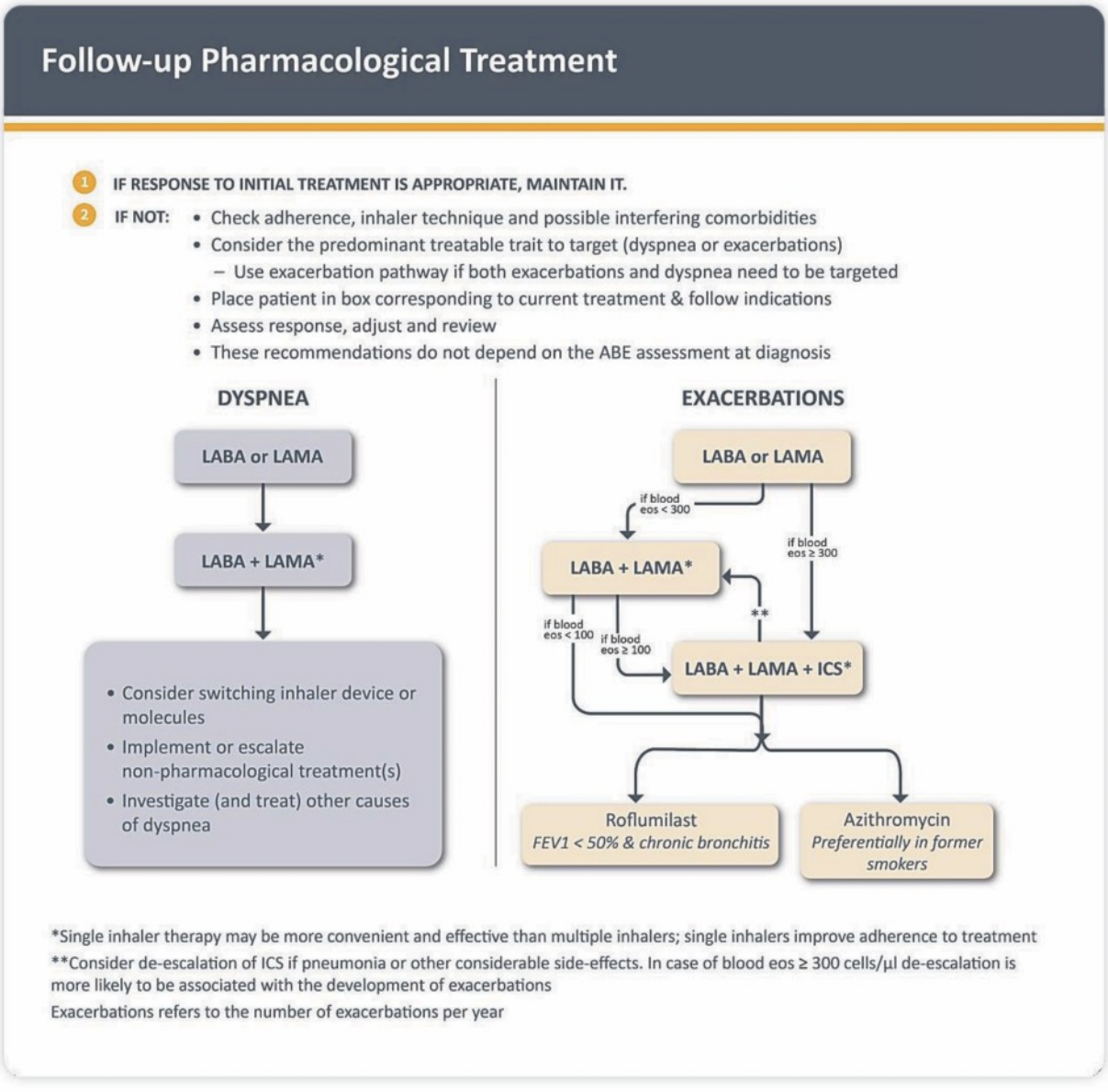
  This is guided by the principles of first ***review*** and ***assess***, then ***adjust*** if necessary :-

- **Review:** Review symptoms (dyspnea) and exacerbation risk (previous history, blood eosinophils).
- **Assess:** Assess inhaler technique and adherence, and the role of non-pharmacological approaches
- **Adjust:** Adjust pharmacological treatment, including escalation or de-escalation. Switching inhaler device or molecules within the same class (e.g., using a different long-acting bronchodilator) may be considered as appropriate. Any change in treatment requires a subsequent ***review*** of the clinical response, including side effects.



Follow-up pharmacological management

The management is based on two key treatable traits: persistence of dyspnea and occurrence of exacerbations. The follow-up recommendations are designed to facilitate management of patients taking maintenance treatment(s), whether early after initial treatment or after years of follow-up.



Dyspnea

- For patients with persistent breathlessness or exercise limitation on ***bronchodilator*** monotherapy, the use oftwo long acting bronchodilators is recommended.
- If the addition of a second long acting bronchodilator does not improve symptoms, switch inhaler device or molecules.
- At all stages, dyspnea due to other causes (not COPD) should be investigated and treated appropriately. Inhaler technique and adherence should be considered as causes of inadequate treatment response. Rehabilitation should also be considered.

Exacerbations

- For patients with persistent exacerbations on ***bronchodilator*** monotherapy, escalation to LABA+LAMA isrecommended.
- In patients who develop further exacerbations on **LABA+LAMA** therapy we suggest escalation to **LABA+LAMA+ICS.**

A beneficial response after the addition of ICS may be observed at blood eosinophil counts ≥ 100 cells/μL, with a greater magnitude of response more likely with higher eosinophil counts.

- If patients treated with **LABA+LAMA+ICS** (or those with eos < 100 cells/μL) still have exacerbations the followingoptions may be considered:
- **Add roflumilast.** This may be considered in patients with an FEV1 < 50% predicted and chronic bronchitis, particularly if they have experienced at least one hospitalization for an exacerbation in theprevious year.
- **Add a macrolide.** The best available evidence exists for the use of azithromycin, especially in those whoare not current smokers. Consideration to the development of resistant organisms should be factored into decision-making.
- **Withdrawing ICS** can be considered if pneumonia or other considerable side-effects develop. If blood eosinophils are ≥ 300 cells/μL de-escalation is more likely to be associated with the development of exacerbations. (Carefully consider the dose of ICS used to reduce the potential of ICS related sideeffects that are more frequent at higher doses.

Patients under treatment with LABA+ICS

- If a patient with COPD and **no features of asthma** has been treated – for whatever reason – with LABA+ICS and iswell controlled in terms of symptoms and exacerbations, continuation with LABA+ICS is an option. However, if the patient has:
- **Further exacerbations:** treatment should be escalated to LABA+LAMA+ICS if the blood eosinophil count is ≥ 100 cells/μL or switched to LABA+LAMA if it is < 100 cells/μL.
- **Major symptoms:** switching to LABA+LAMA should be considered.

Managing inhaled therapy

Most of the drugs used to treat COPD are inhaled. Thus, appropriate use of inhaler devices is crucial to optimize the benefit-risk ratio of inhaled therapy.

Non- Pharmacological treatment of stable COPD

Non-pharmacological treatment is complementary to pharmacological treatment and should form part of the comprehensive management of COPD.

After receiving a diagnosis of COPD a patient should be given further information about the condition. Physicians should emphasize the importance of a smoke free environment, empower adherence to prescribed medication, ensure proper inhaler technique, promote physical activity, prescribe vaccinations, and refer patients to pulmonary rehabilitation.

Algorithms for the initiation and follow-up of non-pharmacological treatment

Some relevant non-pharmacological measures based on the patient's GOLD A,B,E group

Non-Pharmacological Management of COPD*

Patient Group	Essential	Recommended	Depending on Local Guidelines
A	Smoking cessation (can include pharmacological treatment)	Physical activity	Influenza vaccination COVID-19 vaccinations Pneumococcal vaccination Pertussis vaccination Shingles vaccination RSV vaccination
B and E	Smoking cessation (can include pharmacological treatment) Pulmonary rehabilitation	Physical activity	Influenza vaccination COVID-19 vaccinations Pneumococcal vaccination Pertussis vaccination Shingles vaccination RSV vaccination

*Can include pharmacological treatment

Recommendations for **FOLLOW UP** non-pharmacological treatments are based on a patient's treatable traits e.g.,symptoms and exacerbations .

Follow-up of Non-Pharmacological Treatment

1. If response to initial treatment is appropriate, maintain it and offer:

- Influenza vaccination every year and other recommended vaccinations according to guidelines
- Self-management education
- Assessment of behavioral risk factors such as smoking cessation (if applicable) and environmental exposures

Ensure

Maintenance of exercise program and physical activity
Adequate sleep and a healthy diet

2. If not, consider the predominant treatable trait to target

DYSPNEA

- Self-management education (written action plan) with integrated self-management regarding:
  - Breathlessness, energy conservation techniques, and stress management strategies
- Pulmonary rehabilitation (PR) program and/or maintenance exercise program post PR

EXACERBATIONS

- Self-management education (written action plan) that is personalized with respect to:
  - Avoidance of aggravating factors
  - How to monitor/manage worsening of symptoms
  - Contact information in the event of an exacerbation
- Pulmonary rehabilitation (PR) program and/or maintenance exercise program post PR

All patients with advanced COPD should be considered for end of life and palliative care support to optimize symptom control and allow patients and their families to make informed choices about future management.

Rehabilitation, Education & Self-Management

upportive, palliative, end-of-life & hospicePulmonary rehabilitation

Pulmonary rehabilitation is defined as "a comprehensive intervention based on thorough patient assessment followed by patient-tailored therapies that include, but are not limited to, exercise training, education, self-management intervention aiming at behavior change, designed to improve the physical and psychological condition of people with chronic respiratory disease and to promote the long-term adherence to health-enhancing behaviors.

Patients with high symptom burden and risk of exacerbations (Groups B and E), should be encouraged to take part in a formal rehabilitation program that includes setting patient goals and is designed and delivered in a structured manner, taking into account the individual's COPD characteristics and comorbidities. This includes patients who are older, female, more deprived, or have a comorbidity of diabetes, asthma, or painful condition and currently appear less likely to be referred for pulmonary rehabilitation.

Assessment and follow-up of pulmonary rehabilitation

Baseline and outcome assessments of each participant in a pulmonary rehabilitation program should be made to specify individual maladaptive behaviors (including motivation), physical and mental health impediments to training, goals, barriers and capabilities and to quantify gains and to target areas for improvement.

Assessments should include:

- Detailed history and physical examination
- Measurement of post-bronchodilator spirometry
- Assessment of exercise capacity
- Measurement of health status and impact of breathlessness
- Assessment of inspiratory and expiratory muscle strength and lower limb strength in patients who suffer from muscle wasting
- Discussion about individual patient goals and expectations

The first two assessments are important for establishing entry suitability and baseline status but are not used in outcome assessment.

Pulmonary Rehabilitation, Self-Management and Integrative Care in COPD

Pulmonary Rehabilitation	• Pulmonary rehabilitation improves dyspnea, health status and exercise tolerance in stable patients **(Evidence A)** • Pulmonary rehabilitation reduces hospitalization among patients who have had a recent exacerbation (≤ 4 weeks from prior hospitalization) **(Evidence B)** • Pulmonary rehabilitation leads to a reduction in symptoms of anxiety and depression **(Evidence A)**
Education and Self-Management	• Education alone has not been shown to be effective **(Evidence C)** • Self-management intervention with communication with a health care professional improves health status and decreases hospitalizations and emergency department visits **(Evidence B)**
Integrated Care Programs	• Integrative care and telehealth have no demonstrated benefit at this time **(Evidence B)**

Pulmonary Rehabilitation, Self-Management and Integrative Care in COPD

Pulmonary Rehabilitation	• Rehabilitation is indicated in all patients with relevant symptoms and/or a high risk for exacerbation **(Evidence A)** • Pulmonary rehabilitation improves dyspnea, health status and exercise tolerance in stable patients **(Evidence A)** • Pulmonary rehabilitation reduces hospitalization among patients who have had a recent exacerbation (≤ 4 weeks from prior hospitalization) **(Evidence B)** • Pulmonary rehabilitation leads to a reduction in symptoms of anxiety and depression **(Evidence A)**
Education and Self-Management	• Education is needed to change patient's knowledge but there is no evidence that used alone it will change patient behavior **(Evidence C)** • Self-management intervention with communication with a health care professional improves health status and decreases hospitalizations and emergency department visits **(Evidence B)**
Integrated Care Programs	• Integrative care and telehealth have no demonstrated benefit at this time **(Evidence B)**
Physical Activity	• Physical activity is a strong predictor of mortality **(Evidence A)**. People with COPD should be encouraged to increase their level of physical activity although we still do not know how to best ensure the likelihood of success

Oxygen therapy and ventilatory support

The long-term administration of oxygen (> 15 hours per day) to patients with chronic respiratory failure has been shown to increase survival in patients with severe resting hypoxemia.

Oxygen Therapy and Ventilatory Support in Stable COPD

Oxygen Therapy	• The long-term administration of oxygen increases survival in patients with severe chronic resting arterial hypoxemia **(Evidence A)** • In patients with stable COPD and moderate resting or exercise-induced arterial desaturation, prescription of long-term oxygen does not lengthen time to death or first hospitalization or provide sustained benefit in health status, lung function and 6-minute walk distance **(Evidence A)** • Resting oxygenation at sea level does not exclude the development of severe hypoxemia when traveling by air **(Evidence C)**
Ventilatory Support	• NPPV may improve hospitalization-free survival in selected patients after recent hospitalization, particularly in those with pronounced daytime persistent hypercapnia ($PaCO_2$ > 53 mmHg) **(Evidence B)** • In patients with severe chronic hypercapnia and a history of hospitalization for acute respiratory failure, long-term noninvasive ventilation may be considered **(Evidence B)**

Prescription of Supplemental Oxygen to COPD Patients

Arterial hypoxemia defined as:
$PaO_2 \leq 55$ mmHg (7.3 kPa) or $SaO_2 < 88\%$
or
$PaO_2 > 55$ but < 60 mmHg (> 7.3 kPa but < 8 kPa)
with right heart failure or erythrocytosis

↓

Prescribe supplemental oxygen
and titrate to keep $SaO_2 \geq 90\%$

↓

Recheck in 60 to 90 days to assess:
- If supplemental oxygen is still indicated
- If prescribed supplemental oxygen is effective

Ventilatory support

Noninvasive ventilation (NIV) is occasionally used in patients with stable very severe COPD. NIV may be considered of some use in a selected group of patients, particularly in those with pronounced daytime hypercapnia and recent hospitalization. In contrast, in patients with both COPD and obstructive sleep apnea there are clear indications for continuous positive airway pressure (CPAP).

During exacerbations of COPD

NIV in the form of noninvasive positive pressure ventilation (NPPV) is the standard of care for decreasing morbidity and mortality in patients hospitalized with an exacerbation of COPD and acute respiratory failure·

Stable patient

In patients with both COPD and obstructive sleep apnea there are clear benefits associated with the use of continuous positive airway pressure (CPAP) to improve both survival and the risk of hospital admissions.

Whether to use NPPV chronically at home to treat patients with acute on chronic respiratory failure following hospitalization remains undetermined and outcome may be affected by persistent hypercapnia.

Monitoring and follow-up

Routine follow-up of COPD patients is essential. Lung function may worsen over time, even with the best available care. Symptoms, exacerbations and objective measures of airflow obstruction should be monitored to determine when to modify management and to identify any complications and/or comorbidities that may develop.

Symptoms

At each visit, information on symptoms since the last visit should be collected, including cough and sputum, breathlessness, fatigue, activity limitation, and sleep disturbances, including insomnia. Questionnaires such as the COPD Assessment Test (CAT™) can be used.

Exacerbations

The frequency, severity, type and likely causes of all exacerbations should be monitored. Sputum volume and presence or absence of sputum purulence should be noted. Specific inquiry into response to previous treatment, unscheduled visits to providers, telephone calls for assistance, and use of urgent or emergency care facilities is important. Hospitalizations should be documented, including the facility, duration of stay, and any use of critical care or mechanical ventilatory support.

Adherence and appropriate use of prescribed treatments

This is a key action in the chronic management of COPD patients that should be mandatory in each clinical visit. The following aspects need careful and personalized attention:

- Dosages of prescribed medications
- Adherence to the regimen
- Inhaler technique
- Effectiveness of the current regime
- Side effects

Treatment modifications should be recommended.

Smoking status

At each visit, the current smoking status and smoke exposure should be determined followed by appropriate action.

Measurements

Decline in FEV1 can be tracked by spirometry performed at regular intervals (e.g., yearly) to identify patients who are declining quickly, although other lung function parameters reflecting hyperinflation and gas transfer may also be informative.

Imaging

If there is a clear worsening of symptoms, imaging may be indicated. When exacerbations are repeatedly characterized by purulent sputum, patients should be investigated for bronchiectasis.

Comorbidities

Symptoms that may indicate the development or worsening of a comorbid condition such as lung cancer, obstructive sleep apnea, congestive heart failure, ischemic heart disease, osteoporosis or depression/anxiety etc. should be recorded. If present, an appropriate diagnostic work-up should follow.

Supportive, palliative, end-of-life & hospice care

Palliative care is a broad term that encompasses approaches to symptom control as well as management of terminal patients close to death. The goal of palliative care is to prevent and relieve suffering, and to support the best possible quality of life for patients and their families, regardless of the stage of disease or the need for other therapies.
COPD is a highly symptomatic disease and has many elements such as fatigue, dyspnea, depression, anxiety, insomnia that require symptom-based palliative treatments

Palliative Care, End of Life and Hospice Care in COPD

- All clinicians managing patients with COPD should be aware of the effectiveness of palliative approaches to symptom control and use these in their practice **(Evidence D)**
- End of life care should include discussions with patients and their families about their views on resuscitation, advance directives and place of death preferences **(Evidence D)**
- Opiates, neuromuscular electrical stimulation (NMES), oxygen and fans blowing air onto the face can relieve breathlessness **(Evidence C)**
- Nutritional supplementation should be considered in malnourished patients with COPD (**Evidence B**) as it may improve respiratory muscle strength and overall health status **(Evidence B)**
- Fatigue can be improved by self-management education, pulmonary rehabilitation, nutritional support and mind-body interventions **(Evidence B)**

Management of exacerbations

- An exacerbation of COPD is defined as an event characterized by dyspnea and/or cough and sputum that worsen over < 14 days. Exacerbations of COPD are often associated with increased local and systemic inflammation caused by airway infection, pollution, or other insults to the lungs.

- As the symptoms are not specific to COPD relevant differential diagnoses should be considered, particularly pneumonia, congestive heart failure and pulmonary embolism.

- The goals for treatment of COPD exacerbations are to minimize the negative impact of the current exacerbation and to prevent subsequent events.

- Short-acting inhaled beta2-agonists, with or without short-acting anticholinergics, are recommended as the initial bronchodilators to treat an exacerbation.

- Maintenance therapy with long-acting bronchodilators should be initiated as soon as possible. In patients with frequent exacerbations and elevated blood eosinophil levels addition of inhaled corticosteroids to the double bronchodilator regimen should be considered.

- In patients with severe exacerbations, systemic corticosteroids can improve lung function (FEV1), oxygenation and shorten recovery time including hospitalization duration. Duration of therapy should not normally be more than 5 days.

- Antibiotics, when indicated, can shorten recovery time, reduce the risk of early relapse, treatment failure, and hospitalization duration. Duration of therapy should be 5 days.

- Methylxanthines are not recommended due to increased side effect profiles.

- Non-invasive mechanical ventilation should be the first mode of ventilation used in COPD patients with acute respiratory failure who have no absolute contraindication because it improves gas exchange, reduces work of breathing and the need for intubation, decreases hospitalization duration and improves survival.

- Exacerbation recovery time varies, taking up to 4-6 weeks to recover, with some patients failing to return to the pre-exacerbation functional state. Following an exacerbation, appropriate measures for exacerbation prevention should be initiated.

Classification of the Severity of COPD Exacerbations

COPD Patient with Suspected Exacerbation

Confirm ECOPD Diagnosis and Episode Severity

Severity	Variable thresholds to determine severity
Mild (default)	• Dyspnea VAS < 5 • RR < 24 breaths/min • HR < 95 bpm • Resting SaO_2 ≥ 92% breathing ambient air (or patient's usual oxygen prescription) AND change ≤ 3% (when known) • CRP < 10 mg/L (if obtained)
Moderate (meets at least three of five*)	• Dyspnea VAS ≥ 5 • RR ≥ 24 breaths/min • HR ≥ 95 bpm • Resting SaO_2 < 92% breathing ambient air (or patient's usual oxygen prescription) AND/OR change > 3% (when known) • CRP ≥ 10 mg/L *If obtained, ABG may show hypoxemia (PaO_2 ≤ 60 mmHg) and/or hypercapnia ($PaCO_2$ > 45 mmHg) but no acidosis
Severe	• Dyspnea, RR, HR, SaO_2 and CRP same as moderate • ABG show new onset/worsening hypercapnia and acidosis ($PaCO_2$ > 45 mmHg and pH <7.35)

Consider Differential Diagnosis

- Heart failure
- Pneumonia
- Pulmonary embolism

- Appropriate testing and treatment

Determine etiology:
viral testing, sputum culture, other

Adapted from: The ROME Proposal, Celli et al. (2021) Am J Respir Crit Care Med. 204(11): 1251-8.

Abbreviations: VAS visual analog dyspnea scale; RR respiratory rate; HR heart rate; SaO_2 oxygen saturation; CRP C-reactive protein; ABG arterial blood gases; PaO_2 Arterial pressure of oxygen.

Management of Severe but not Life-threatening Exacerbations*

Assess severity of symptoms, blood gases, chest radiograph

Administer supplemental oxygen therapy, obtain serial arterial blood gas, venous blood gas and pulse oximetry measurements

Bronchodilators:

- Increase doses and/or frequency of short-acting bronchodilators
- Combine short-acting beta 2-agonists and anticholinergics
- Consider use of long-acting bronchodilators when patient becomes stable
- Use spacers or air-driven nebulizers when appropriate

Consider oral corticosteroids

Consider antibiotics (oral) when signs of bacterial infection are present

Consider noninvasive mechanical ventilation (NIV)

At all times:

- Monitor fluid balance
- Consider subcutaneous heparin or low molecular weight heparin for thromboembolism prophylaxis
- Identify and treat associated conditions (e.g., heart failure, arrhythmias, pulmonary embolism etc.)

*Local resources need to be considered

Key Points for the Management of Exacerbations

- Short-acting inhaled beta$_2$-agonists, with or without short-acting anticholinergics, are recommended as the initial bronchodilators to treat an acute exacerbation **(Evidence C)**
- Systemic corticosteroids can improve lung function (FEV1), oxygenation and shorten recovery time and hospitalization duration. Duration of therapy should not normally be more than 5 days **(Evidence A)**
- Antibiotics, when indicated, can shorten recovery time, reduce the risk of early relapse, treatment failure, and hospitalization duration. Duration of therapy should normally be 5 days **(Evidence B)**
- Methylxanthines are not recommended due to increased side effect profiles **(Evidence B)**
- Non-invasive mechanical ventilation should be the first mode of ventilation used in COPD patients with acute respiratory failure who have no absolute contraindication because it improves gas exchange, reduces work of breathing and the need for intubation, decreases hospitalization duration and improves survival **(Evidence A)**

Interventions that Reduce the Frequency of COPD Exacerbations

Intervention Class	Intervention
Bronchodilators	LABAs LAMAs LABA + LAMA
Corticosteroid-containing regimens	LABA + ICS LABA + LAMA + ICS
Anti-inflammatory (non-steroid)	Roflumilast
Anti-infectives	Vaccines Long Term Macrolides
Mucoregulators	N-acetylcysteine Carbocysteine Erdosteine
Various others	Smoking Cessation Rehabilitation Lung Volume Reduction Vitamin D Shielding measures (e.g., mask wearing, minimizing social contact, frequent hand washing)

Common Risk Factors for the Development of Lung Cancer

- Age > 55 years
- Smoking history > 30 pack years
- Presence of emphysema by CT scan
- Presence of airflow limitation FEV1/FVC < 0.7
- BMI < 25 kg/m^2
- Family history of lung cancer

COPD and Comorbidities

- COPD often coexists with other diseases (comorbidities) that may have a significant impact on disease course.

- In general, the presence of comorbidities should not alter COPD treatment and comorbidities should be treated per usual standards regardless of the presence of COPD.

- Cardiovascular diseases are common and important comorbidities in COPD.

- Lung cancer is frequently seen in people with COPD and is a major cause of death.
  - Annual low-dose CT scan (LDCT) is recommended for lung cancer screening in people with COPD due to smoking according to recommendations for the general population

  - Annual LDCT is not recommended for lung cancer screening in people with COPD not due to smoking due to insufficient data to establish benefit over harm

- Osteoporosis and depression/anxiety are frequent, important comorbidities in COPD, are often under-diagnosed, and are associated with poor health status and prognosis.

- Gastroesophageal reflux (GERD) is associated with an increased risk of exacerbations and poorer health status.

- When COPD is part of a multimorbidity care plan, attention should be directed to ensure simplicity of treatment and to minimize polypharmacy.

Key Points for the Management of Stable COPD During COVID-19 Pandemic

Protective Strategies	• Follow basic infection control measures • Wear a face covering • Consider shielding/sheltering-in-place • Have the COVID-19 vaccinations in line with national recommendations
Investigations	• Only essential spirometry at times of high prevalence of COVID-19
Pharmacotherapy	• Ensure adequate supplies of medications • Continue unchanged including ICS
Non-Pharmacological Therapy	• Ensure annual influenza vaccination • Maintain physical activity

References

1. Global initiative for chronic obstructive lung disease pocket guide to COPD diagnosis, management, and prevention a guide for health care professionals, 2024 report.
2. Diagnosis and management of COPD in primary care. A guide for those working in primary care. Publication reference 1001, 2007, Edition 6 (August 2015).
3. COPD action plan, the American lung association: - available from https://www.lung.org/assets/documents/copd/copd-action-plan.pdf
4. Air pollution and COPD: GOLD 2023
   Don D. Sin, Dany Doiron, Alvar Agusti, Antonio Anzueto, Peter J. Barnes, Bartolome R. Celli, Gerard J. Criner, David Halpin, MeiLan K. Han, Fernando J. Martinez, Maria Montes de Oca, Alberto Papi, Ian Pavord, Nicolas Roche, Dave Singh, Robert Stockl ey, M. Victorina Lopez Varlera, Jadwiga Wedzicha, Claus Vogelmeier, Jean Bourbeau on behalf of the GOLD Scientific Committee
   European Respiratory Journal 2023 61: 2202469; DOI: 10.1183/13993003.02469-2022

ANNEXEX

Bronchodilators in Stable COPD

- Inhaled bronchodilators in COPD are central to symptom management and commonly given on a regular basis to prevent or reduce symptoms **(Evidence A)**
- Inhaled bronchodilators are recommended over oral bronchodilators (**Evidence A**)
- Regular and as-needed use of SABA or SAMA improves FEV1 and symptoms **(Evidence A)**
- Combinations of SABA and SAMA are superior compared to either medication alone in improving FEV1 and symptoms **(Evidence A)**
- LABAs and LAMAs are preferred over short-acting agents except for patients with only occasional dyspnea (**Evidence A**), and for immediate relief of symptoms in patients already on long-acting bronchodilators for maintenance therapy
- LABAs and LAMAs significantly improve lung function, dyspnea, health status, and reduce exacerbation rates **(Evidence A)**
- LAMAs have a greater effect on exacerbation reduction compared with LABAs **(Evidence A)** and decrease hospitalizations **(Evidence B)**
- When initiating treatment with long acting bronchodilators the preferred choice is a combination of a LABA and a LAMA. In patients with persistent dyspnea on a single long-acting bronchodilator treatment should be escalated to two (**Evidence A**).
- Combination treatment with a LABA and a LAMA increases FEV1 and reduces symptoms compared to monotherapy **(Evidence A)**
- Combination treatment with a LABA+LAMA reduces exacerbations compared to monotherapy **(Evidence B)**
- Combinations can be given as single inhaler or multiple inhaler treatment. Single inhaler therapy may be more convenient and effective than multiple inhalers
- Theophylline exerts a small bronchodilator effect in stable COPD **(Evidence A)** and that is associated with modest symptomatic benefits **(Evidence B)**

Commonly Used Maintenance Medications in COPD*

Generic Drug Name	Inhaler Type	DELIVERY OPTIONS			Duration of Action
		Nebulizer	Oral	Injection	
$BETA_2$-Agonists					
Short-acting (SABA)					
Fenoterol	MDI	✓	pill, syrup		4-6 hours
Levalbuterol	MDI	✓			6-8 hours
Salbutamol (albuterol)	MDI & DPI	✓	pill, syrup, extended release tablet	✓	4-6 hours 12 hours (ext. release)
Terbutaline	DPI		pill	✓	4-6 hours
Long-acting (LABA)					
Arformoterol		✓			12 hours
Formoterol	DPI	✓			12 hours
Indacaterol	DPI				24 hours
Olodaterol	SMI				24 hours
Salmeterol	MDI & DPI				12 hours
Anticholinergics					
Short-acting (SAMA)					
Ipratropium bromide	MDI	✓			6-8 hours
Oxitropium bromide	MDI				7-9 hours
Long-acting (LAMA)					
Aclidinium bromide	DPI				MDI 12 hours
Glycopyrronium bromide	DPI		solution	✓	12-24 hours
Tiotropium	DPI, SMI, MDI				24 hours
Umeclidinium	DPI				24 hours
Glycopyrronium		✓			12 hours
Revefenacin		✓			24 hours
Combination Short-Acting $Beta_2$-Agonist Plus Anticholinergic in One Device (SABA+SAMA)					
Fenoterol/ipratropium	SMI	✓			6-8 hours
Salbutamol/ipratropium	SMI, MDI	✓			6-8 hours
Combination Long-Acting $Beta_2$-Agonist Plus Anticholinergic in One Device (LABA+LAMA)					
Formoterol/aclidinium	DPI				12 hours
Formoterol/glycopyrronium	MDI				12 hours
Indacaterol/glycopyrronium	DPI				12-24 hours
Vilanterol/umeclidinium	DPI				24 hours
Olodaterol/tiotropium	SMI				24 hours
Methylxanthines					
Aminophylline			solution	✓	Variable, up to 24 hours
Theophylline (SR)			pill	✓	Variable, up to 24 hours
Combination of Long-Acting $Beta_2$-Agonist Plus Corticosteroid in One Device (LABA+ICS)					
Formoterol/beclometasone	MDI, DPI				12 hours
Formoterol/budesonide	MDI, DPI				12 hours
Formoterol/mometasone	MDI				12 hours
Salmeterol/fluticasone propionate	MDI, DPI				12 hours
Vilanterol/fluticasone furoate	DPI				24 hours
Triple Combination in One Device (LABA+LAMA+ICS)					
Fluticasone/umeclidinium/vilanterol	DPI				24 hours
Beclometasone/formoterol/glycopyrronium	MDI, DPI				12 hours
Budesonide/formoterol/glycopyrrolate	MDI				12 hours
Phosphodiesterase-4 Inhibitors					
Roflumilast			pill		24 hours
Mucolytic Agents					
Erdosteine			pill		12 hours
Carbocysteine†			pill		
N-acetylcysteine†			pill		

*Not all formulations are available in all countries. In some countries other formulations and dosages may be available. †Dosing regimens are under discussion.
MDI = metered dose inhaler; DPI = dry powder inhaler; SMI = soft mist inhaler. Note that glycopyrrolate & glycopyrronium are the same compound.

Anti-Inflammatory Therapy in Stable COPD

Inhaled Corticosteroids	• Regular treatment with ICS increases the risk of pneumonia especially in those with severe disease **(Evidence A)** • An ICS combined with a LABA is more effective than the individual components in improving lung function and health status and reducing exacerbations in patients with exacerbations and moderate to very severe COPD **(Evidence A)** • We do not encourage the use of a LABA+ICS combination in COPD. If there is an indication for an ICS the combination LABA+LAMA+ICS has been shown to be superior to LABA+ICS and is therefore the preferred choice • Triple inhaled therapy of LABA+LAMA+ICS improves lung function, symptoms and health status, and reduces exacerbations, compared to LABA+ICS, LABA+LAMA or LAMA monotherapy **(Evidence A)**. Recent data suggesta beneficial effect of triple inhaled therapy versus fixed-dose LABA+LAMA combinations on mortality in symptomatic COPD patients with a history of frequent and/or severe exacerbations • If patients with COPD have features of asthma, treatment should always contain an ICS • Independent of ICS use, there is evidence that a blood eosinophil count < 2% increases the risk of pneumonia **(Evidence C)** • Combinations can be given as single or multiple inhaler therapy. Single inhaler therapy may be more convenient and effective than multiple inhalers
Oral Glucocorticoids	• Long-term use of oral glucocorticoids has numerous side effects **(Evidence A)** with no evidence of benefits **(Evidence C)**
PDE4 Inhibitors	• In patients with chronic bronchitis, severe to very severe COPD and a history of exacerbations: ▪ Roflumilast improves lung function and reduces moderate and severe exacerbations **(Evidence A)**
Antibiotics	• Long-term azithromycin and erythromycin therapy reduces exacerbations over one year **(Evidence A)** • Preferentially, but not only in former smokers with exacerbations despite appropriate therapy, azithromycin can be considered (**Evidence B**) • Treatment with azithromycin is associated with an increased incidence of bacterial resistance **(Evidence A)** and hearing test impairments **(Evidence B)**
Mucoregulators and Antioxidant Agents	• Regular treatment with mucolytics such as erdosteine, carbocysteine and NAC reduces the risk of exacerbations in select populations **(Evidence B)** • Antioxidant mucolytics are recommended only in selected patients **(Evidence A)**
Other Anti-Inflammatory Agents	• Statin therapy is not recommended for prevention of exacerbations (**Evidence A**) • Simvastatin does not prevent exacerbations in COPD patients at increased risk of exacerbations and without indications for statin therapy **(Evidence A)**. However, observational studies suggest that statins may have positive effects on some outcomes in patients with COPD who receive them for cardiovascular and metabolic indications **(Evidence C)** • Leukotriene modifiers have not been tested adequately in COPD patients

Factors to Consider when Initiating ICS Treatment

Factors to consider when adding ICS to long-acting bronchodilators:

(note the scenario is different when considering ICS withdrawal)

STRONGLY FAVORS USE	History of hospitalization(s) for exacerbations of COPD#
	≥ 2 moderate exacerbations of COPD per year#
	Blood eosinophils ≥ 300 cells/μL
	History of, or concomitant asthma
FAVORS USE	1 moderate exacerbation of COPD per year#
	Blood eosinophils 100 to < 300 cells/μL
AGAINST USE	Repeated pneumonia events
	Blood eosinophils < 100 cells/μL
	History of mycobacterial infection

#despite appropriate long-acting bronchodilator maintenance therapy (see Figures 3.7 & 3.18 for recommendations); *note that blood eosinophils should be seen as a continuum; quoted values represent approximate cut-points; eosinophil counts are likely to fluctuate.

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Other Pharmacological Treatments

Alpha-1 Antitrypsin Augmentation Therapy	• Intravenous augmentation therapy may slow down the progression of emphysema **(Evidence B)**
Antitussives	• There is no conclusive evidence of a beneficial role of antitussives in people with COPD **(Evidence C)**
Vasodilators	• Vasodilators do not improve outcomes and may worsen oxygenation **(Evidence B)**
Opioids	• Low-dose long acting oral and parenteral opioids may be considered for treating dyspnea in COPD patients with severe disease **(Evidence B)**
Pulmonary Hypertension Therapy	• Drugs approved for primary pulmonary hypertension are not recommended for patients with a pulmonary hypertension secondary to COPD **(Evidence B)**

COPD FOLLOW-UP CHECKLIST

In-person Follow-up **Phone Follow-up** **Virtual/online Follow-up** □

Date: / /	**Diagnosis:**

1. BASELINE SYMPTOMS – Breathlessness on a regular day: mMRC / 4

Daily sputum production: □ no □ yes, color: Regular cough □ no □ yes

Recent change in symptoms □ no □ yes If yes, since when:	*Maintenance Medication and adherence:*
□ Sputum color: □ Sputum volume ↑ = ↓	□ SABA □ LABA+LAMA
□ Dyspnea ↑ = ↓ □ Fatigue ↑ = ↓	□ LABA □ LABA+ICS
□ Cough ↑ = ↓ □ Other	□ LAMA □ LABA+LAMA+ICS
□ Signs of hypercapnia **CAT: /40**	□Other:
	Non pharmacological Rx:
	O2: CPAP: BIPAP :

2. COVID-19 – If patient is feeling unwell, check other symptoms: □ Fever_____□ Sore throat □ Anosmia □ Others _____ Contact with someone COVID-19 positive? □ no □ yes

Tested for COVID-19? □ no □ yes If yes □ positive □ negative

3. WRITTEN ACTION PLAN – no □ yes □

Instruction and any additional treatment: _____ Last time it has been used (date):

4. RECENT ADMISSIONS AND EMERGENCY VISITS

Hospital/ER	Where	Date	Length	Reason (Dx)	*Comments:*

5. COPD Self-management (healthy behaviors) – Integrated (patient has used it in his daily life)**?**

Smoke-free environment yes no cannot tell
Medication adherence yes no cannot tell Prevention/management of exacerbations yes no cannot tell Breathing control yes no cannot tell
Stress management yes no cannot tell
Physical activity and exercise yes no cannot tell
Other _____ yes no
Comments and what patient should prioritize based on his/her need:

6. MAIN ISSUES

1.	*2.*	*3.*

7. SUMMARY, INTERVENTIONS & PLAN